Dependence among organisms

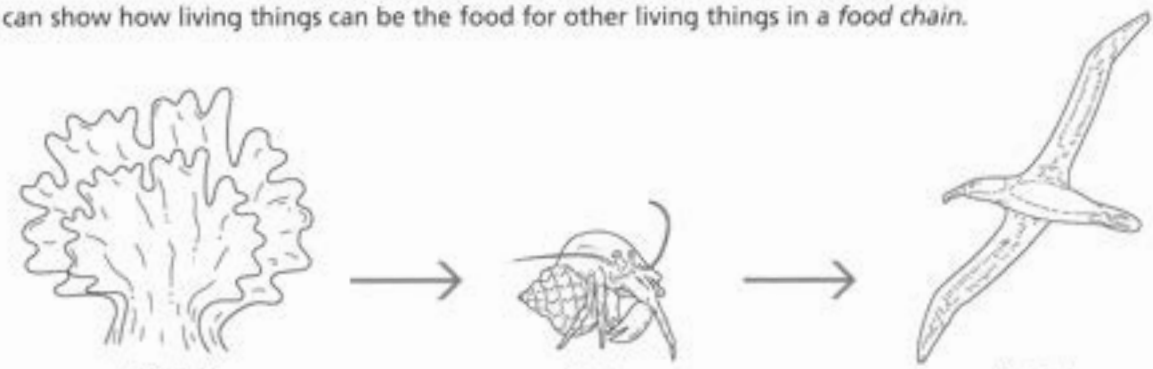
All living things need energy to grow, move, reproduce etc. This energy is provided by food. Plants make their own food using the energy from sunlight in a process called *photosynthesis*.

1 Name one use of energy by organisms not already mentioned.

2 What is the type of energy that comes from the sun?

Green plants are called *producers* because they make their own food. Animals cannot make their own food. They get their energy by eating (consuming) plants or other animals. If there were no plants to eat, animals could not survive. We also call organisms that produce their own food *autotrophs* and those that do not produce their own food are called *heterotrophs*.

We can show how living things can be the food for other living things in a *food chain*.



Seaweed → Crab → Seabird

The seaweed (a form of algae) is a producer. The crab (a crustacean) is a **first-order** consumer because it eats the seaweed. The seabird is a **second-order** consumer because it eats crabs.

3 The seaweed is a marine organism. What is meant by *marine*?

4 If this food chain were extended to include a third-order consumer, what would this third-order consumer be likely to eat?

Food chains show how energy passes from producers to consumers as members of the chain are eaten by other members further up the chain. As each food is digested, energy is released.

5 Does the food chain above show all the organisms involved? Explain your answer.

6 What is meant by the arrows shown in the food chain?

7 The organisms in a food chain below are not in the right order. Write them in the correct order.

Seal	Seaweed	Lobster	Small shellfish

_____ → _____ → _____ → _____

8 Give an example of a *crustacean*.

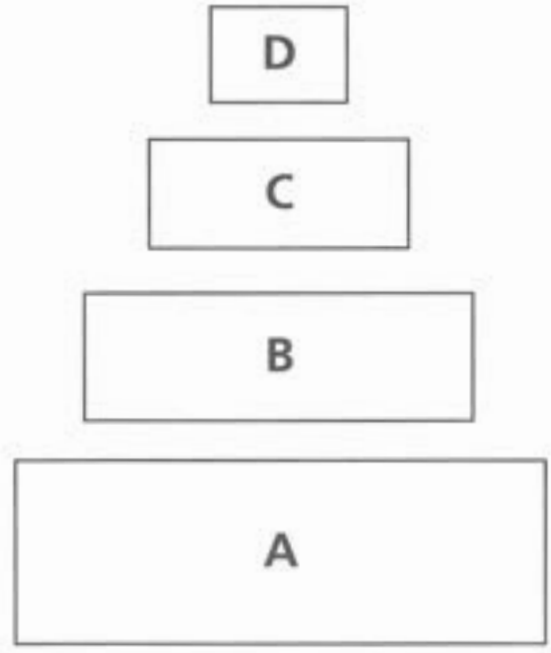
9 Outline the difference between a *heterotroph* and an *autotroph*.

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Food pyramids

As you move further up a food chain, the consumers become larger in size but fewer in number. The amount of *biomass* (biological mass or mass of living things) decreases as you go up the food chain.



In the model of a food pyramid to the right, **A** represents leaves and vegetation, **B** represents insects, **C** represents small birds and **D** represents raptors (birds of prey, eg eagles and hawks.)

Green plants make their own food from solar energy, carbon dioxide and water. When an insect eats a leaf, it uses up some of the energy in moving, growing etc. This means that when the insect is eaten by a small bird not all of the energy is passed to the bird from the leaf because some of it was incorporated into the insect. Therefore the bird needs to eat many insects to get enough food to support its needs. Consequently there will be many more leaves than insects and many more insects than small birds. There will also be fewer birds of prey than small birds.

10 Write out the food chain described in the food pyramid above.

11 Name the producer.

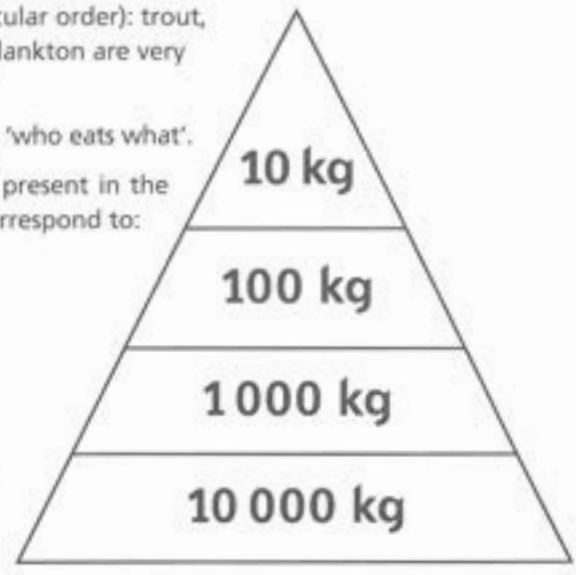
12 Name the third-order consumer.

13 What is a *raptor*?

The following organisms are present in a small lake (in no particular order): trout, insect larvae, fingerlings (small fish) and phytoplankton. Phytoplankton are very small plants that have chlorophyll and can photosynthesise.

14 Write out a food chain for the organisms in the lake, showing 'who eats what'.

The diagram on the right shows the masses of each organism present in the lake. Give the names of the organisms that are most likely to correspond to:



15 10 000 kg

16 1 000 kg

17 100 kg

18 10 kg

A person came along and started to fish in the lake. Eventually she caught all of the trout present.

19 Forecast what might happen to the number of fingerlings. Explain your answer.

20 What are *phytoplankton*?

21 What is *biomass*?

22 There are some organisations in Australia that release millions of young fish into water systems each year. Give a reason why they might do this.

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